

INMO(8th) – 1993

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. The diagonals AC and BD of a cyclic quadrilateral ABCD intersect at P. Let O be the circumcentre of triangle APB and H be the orthocenter of triangle CPD. Show that the points H, P, O are collinear.
 2. Let $P(x) = x^2 + ax + b$ be a quadratic polynomial in which a and b are integers. Given any integers n, show that there is an integer M such that $P(n).P(n+1) = P(M)$.
 3. If a, b, c, d are 4 non-negative real numbers and $a + b + c + d = 1$, show that $ab + bc + cd \leq \frac{1}{4}$.
 4. Let ABC be a triangle in a plane Σ . Find the set of all points P (distinct from A, B, C) in the plane Σ such that the circum circles of triangles ABP, BCP and CAP have the same radii.
 5. Show that there is a natural number n such that n! when written in decimal notation (that in base 10) ends exactly in 1993 zeros.
 6. Let ABC be a triangle right angled at A and S be the circum circle. Let S_1 be the circle touching the lines AB and AC and the circle S internally. Further let S_2 be the circle touching the lines AB and AC, and the circle S externally. If r_1 and r_2 be the radius of the circles S_1 and S_2 respectively, show that $r_1 \cdot r_2 = 4(\text{area} \Delta ABC)$.
 7. Let $A = \{1, 2, 3, \dots, 100\}$ and B be the subset of A having 53 elements. Show that B has two distinct elements x and y whose sum is divisible by 11.
 8. Let f be a bijective (1-1 and onto) function from $A = \{1, 2, 3, \dots, n\}$ to itself. Show that there is positive number $M \geq 1$ such that $f^M(i) = f(i)$ for each I in A. $f^M = \underbrace{f \circ f \circ \dots \circ f}_{M\text{-times}}$ composite function.
 9. Show that there exist a convex hexagon in the plane such that:
 - (a) All its interior angles are equal.
 - (b) All its sides are 1, 2, 3, 4, 5, 6 in some order.

 **